

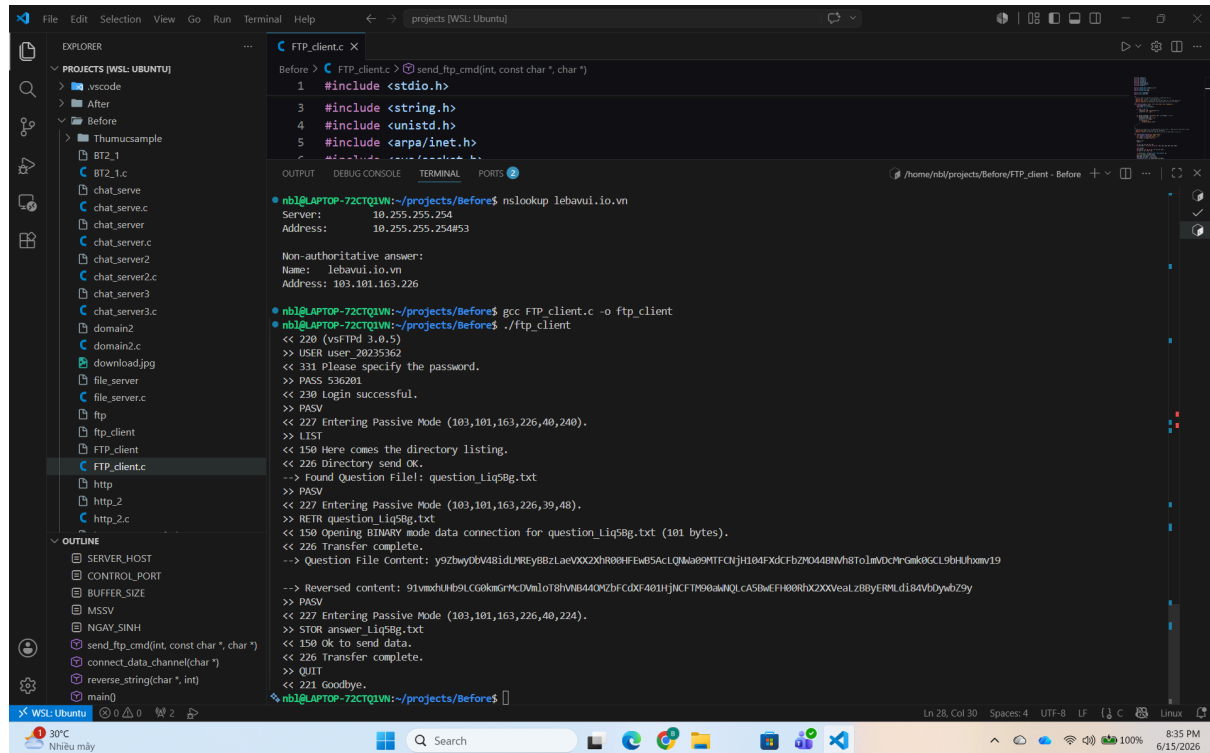
MSSV: 20235362

Ngày sinh: 01/03/2005

Họ và tên: Nguyễn Bảo Linh

BÀI TẬP NGÀY 09/06/2026 - LẬP TRÌNH MẠNG

Bài 1: Lập trình FTP Client:



```
Before > C:FTP_client.c > send_ftplibcmd(int, const char *, char *)
1 #include <stdio.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <arpa/inet.h>

OUTPUT DEBUG CONSOLE TERMINAL PORTS
/home/nbl/projects/Before/FTP_client - Before + - - - - -
nbl@LAPTOP-72CTQ1WN:~/projects/Before$ nslookup lebavui.io.vn
Server: 10.255.255.254
Address: 10.255.255.254#53

Non-authoritative answer:
Name: lebavui.io.vn
Address: 103.101.163.226

nbl@LAPTOP-72CTQ1WN:~/projects/Before$ gcc FTP_client.c -o ftp_client
nbl@LAPTOP-72CTQ1WN:~/projects/Before$ ./ftp_client
<< 220 (vSFTRD 3.0.5)
>> USER user_20235362
<< 331 Please specify the password.
>> PASS 536201
<< 230 Login successful.
>> PASV
<< 227 Entering Passive Mode (103,101,163,226,40,248).
>> LIST
<< 150 Here comes the directory listing.
>> 226 Directory send OK.
--> Found Question File: question_Liq58g.txt
>> PASV
<< 227 Entering Passive Mode (103,101,163,226,39,48).
>> RETR question_Liq58g.txt
<< 150 Opening BINARY mode data connection for question_Liq58g.txt (101 bytes).
<< 226 Transfer complete.
--> Question File Content: y9Zhwj0bV48idUPREY8BzLaeVXXZXR08HFEwB5AclQMa9MIFCjh1B4FXGcfbZM44BNVH8ToImVdCPrGmk9GCL9bHhxm19
--> Reversed content: 91vmxhUHb9LcG0kmGrMcDVmloT8hVNB44OMZbFCdXF401HjNCFTM90aWNQLcA5BwEFH00RhX2XXVealZBBYERMLdi84VbDywbZ9y
>> PASV
<< 227 Entering Passive Mode (103,101,163,226,40,224).
>> STOR answer_Liq58g.txt
<< 150 Ok to send data.
<< 226 Transfer complete.
>> QUIT
<< 221 Goodbye.
nbl@LAPTOP-72CTQ1WN:~/projects/Before$
```

Hình 1. Minh họa chương trình trên terminal

Thứ tự:

- Đăng nhập vào FTP Server lebavui.io.vn
- Lấy danh sách file.
- Tải nội dung file.
- Tạo file answer với nội dung file đảo ngược.
- Upload file lên server.

Nội dung file đảo ngược:

91vmxhUHb9LcG0kmGrMcDVmloT8hVNB44OMZbFCdXF401HjNCFTM90aWNQLcA5BwEFH00RhX2XXVealZBBYERMLdi84VbDywbZ9y